

04 · Which funnel stage to invest in? — mediation (pathmc)

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The business decision. We have a fixed budget to split across the growth funnel — **onboarding** -> **engagement** -> **activation** -> **conversion** — and want to put it where it moves paid conversion the most. The complication is that the stages form a **chain**: money spent improving onboarding doesn't only help onboarding, it *cascades* downstream into engagement, activation, and ultimately conversion. So we can't just correlate each stage with revenue; we have to trace the causal *paths* and decide where a push travels furthest.

1.0.1 The concept: mediation, direct and indirect effects

When a treatment X affects an outcome Y *through* an intermediate variable M (a **mediator**), the total effect splits in two:

- the **direct effect** — X 's influence on Y that does *not* go through M (Pearl's *natural direct effect*, NDE);
- the **indirect effect** — the part that flows $X \rightarrow M \rightarrow Y$ (the *natural indirect effect*, NIE).

Total effect = direct + indirect, and the **proportion mediated** = indirect / total tells you how much of a stage's value is really realised *downstream*. In a linear model the indirect effect is just the **product of the coefficients along the path** (the classic " $a \cdot b$ " of mediation analysis) — **pathmc** generalises this by pushing posterior draws through the intervened graph, so we get every path's effect *with uncertainty*.

1.0.2 The catch this notebook is honest about

Mediation leans on a **stronger** assumption than a simple treatment-effect analysis: no unmeasured confounding of the *mediator-outcome* link (on top of the usual treatment-outcome one). If some unmeasured trait drives both engagement *and* conversion, the direct/indirect *split* is biased even when the *total* effect is fine. So Step 6 doesn't just report the split — it **stress-tests how much such confounding would have to exist to flip the "invest here" decision**.

On real data. Swap in your **product-analytics event data** — per-user onboarding score, engagement, activation flag, conversion — which every growth team already logs. The method needs the funnel's causal *order* to be known (it usually is) and the no-hidden-mediator-confounder assumption to be defensible (Step 6 shows how to argue it).

pathmc builds the chain as a structural causal model and computes **path-specific effects** by propagating posterior draws through the intervened graph — the modern, uncertainty-aware mediation

analysis. 7-step contract.

1.1 2 · Simulate a ground truth

We simulate the funnel as a **causal chain**: `onboarding_score -> engagement -> activated -> converted`, with `channel_quality` feeding both engagement and activation (a common driver we'll need to account for). Each arrow is a real structural equation with a planted coefficient, so we know the **true** total effect of onboarding on conversion *and how it splits across paths* — which is exactly what we'll ask the model to recover. By construction most of onboarding's effect flows the long way round (`engagement -> activated -> converted`), with a smaller slice going `engagement -> converted` directly. If the method can rediscover that split from data alone, we can trust it on a funnel where we *don't* know the answer.

TRUE effects of onboarding on conversion:

indirect_via_engagement_activated	3.17
indirect_via_engagement_only	0.77
total	3.94

	onboarding_score	channel_quality	engagement	activated	converted
0	0.864798	0.726815	2.576471	3.869429	4.870321
1	0.855303	0.790067	3.206651	5.673260	6.371102
2	0.811023	0.348493	2.493785	4.691864	5.394113
3	0.261446	0.465793	0.572682	1.261547	1.264003
4	0.077199	0.887974	1.520775	3.259916	3.146544

1.2 3 · Identify — direct vs indirect effects (NDE / NIE)

For a chain $X \rightarrow M \rightarrow Y$ the **total effect** decomposes into a **natural direct effect** and **natural indirect effect** (Pearl):

$$\text{TE} = \text{NDE} + \text{NIE}, \quad \text{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \quad \text{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))].$$

Under a linear SCM these are the **product-of-coefficients**: indirect = $a \cdot b$, direct = c . The **proportion mediated** = NIE/TE tells you how much of onboarding's value is “really” downstream engagement/activation. `pathmc` computes path-specific effects for the full chain by simulating through the intervened graph.

Assumptions (stronger than a single-link analysis): no unmeasured confounding of $X-Y$, $M-Y$, and $X-M$, plus the **cross-world** condition that no $M-Y$ confounder is itself affected by X . Natural effects are *not identified* if that last one fails — and the $M-Y$ confounding assumption is the one most likely to break, so we stress-test it in step 6.

1.3 4 · Estimate — fit the chain and query every path

We hand `pathmc` the **whole chain at once** — one equation per stage — rather than three disconnected regressions. That matters: because the model knows the full graph, it can propagate an intervention on onboarding *through* engagement and activation to conversion, giving genuine path-specific effects instead of naive stage-by-stage correlations. The `effects_summary()` below

lists every structural coefficient (the a 's and b 's from Step 3); Step 5 then combines them into total, direct, and indirect effects with uncertainty.

	mean	sd	hdi_3%	hdi_97%
name				
e_on	2.150	0.042	2.073	2.226
e_ch	1.131	0.043	1.043	1.207
a_eng	1.616	0.018	1.584	1.652
a_ch	0.592	0.054	0.490	0.690
c_act	0.896	0.022	0.857	0.940
c_eng	0.361	0.040	0.286	0.430

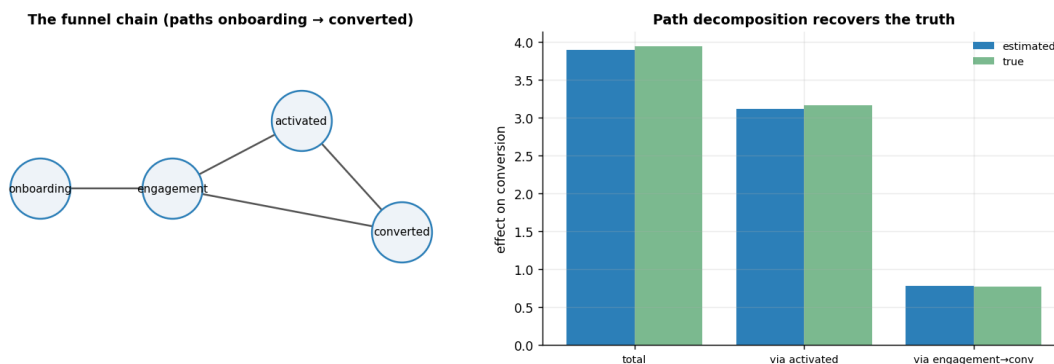
1.4 5 · Validate — recover the total, the paths, and the proportion mediated

We query three things and check each against the planted truth: the **total** effect of onboarding on conversion; the effect flowing through each **path**; and the **proportion mediated** (indirect $\div$ total), which is the business-relevant summary — “what fraction of onboarding’s value is realised *downstream* in engagement and activation, versus directly?” A proportion near 100% means onboarding works almost entirely by feeding the rest of the funnel, which is a very different investment story from a direct effect.

```

TOTAL onboarding -> converted          3.89 (true 3.94) [90% 3.74, 4.04]
  via engagement->activated->conv      3.11 (true 3.17) [90% 2.95, 3.28]
  via engagement->converted             0.78 (true 0.77) [90% 0.63, 0.92]
Proportion mediated (indirect/total): 100% [99%, 101%] - most of onboarding's
value is realised downstream.

```



How to read this. *Left* is the causal graph we fit — note there is **no direct arrow** from onboarding to conversion, so onboarding can only work *through* the funnel (which is why the proportion mediated above came out near 100%). *Right* — the estimated total and each path effect (blue) sit on top of the planted truths (green): the model correctly attributes the bulk of onboarding’s impact to the long **engagement** → **activated** → **converted** route and only a sliver to the direct-from-engagement path. This is the decomposition a naive “regress conversion on each stage” analysis cannot give you.

1.5 6 · Decide, in euros — leverage ranking *and* its fragility

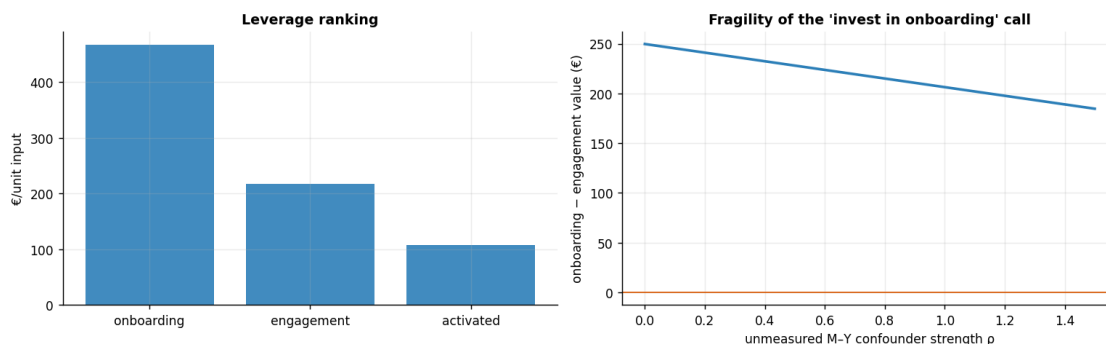
Two moves, and the second is the honest one most mediation analyses skip.

(1) Rank the levers. We value a one-unit improvement in each stage’s input by its **total downstream effect** on conversion x € per conversion. Using the *total* effect (not just the local one) is the whole point — a euro into onboarding cascades through the funnel, so its leverage includes everything it triggers.

(2) Stress-test the split. The direct/indirect decomposition rests on *no unmeasured confounding of the mediator–outcome link* (Step 3). If some omitted trait drives both engagement and conversion, the indirect path is over-credited. So we sweep a hypothetical confounder strength ρ and show **how strong it would have to be to flip the “invest in onboarding vs engagement” decision** — turning an untestable assumption into a concrete “is a confounder that strong plausible here?” question a stakeholder can actually judge.

lever	€/unit	P(>0)
onboarding	467.0	1.0
engagement	217.0	1.0
activated	107.0	1.0

Onboarding leads on leverage, and the call is ROBUST: even the strongest M–Y confounding we swept ($\rho=1.5$) doesn't close the €250 gap to engagement.



1.6 7 · Caveats

- **Natural effects lean on strong assumptions** — no unmeasured confounding of any of the three links plus the cross-world condition. Step 6 quantified how much M–Y confounding would flip the decision; the point estimate alone hides that fragility.
- **Don’t control away the mechanism.** The goal is to *decompose* the path; adjusting for a mediator when you wanted the total effect removes the very thing you’re pricing (nb 05).
- **Linear-SCM decomposition is exact only if the SCM is linear.** For strong nonlinearities lean on `pathmc`’s simulate-through-the-graph numbers, not hand-multiplied coefficients.